

CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

SLAC:

AN ALGORITHM FOR LOSSLESS AUDIO COMPRESSION

Graduate Project in partial fulfillment
of the requirements for the degree
Masters of Science in Computer Science

by

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ABSTRACT

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Lossless compression happens when a pattern can be described with symbols that are only as big as needed at a given instant to uniquely and exactly represent each value. Signals are a special case, as the number of bits needed to represent a value are often very high, and also have the property that each symbol is closely related to the next. Signal compression algorithms go an extra step to find the similarities between adjacent symbols and only store the differences between what the real value is and what some prediction scheme says the value should be.

Typically, compression algorithms examine symbols in a data stream each in whole—whether those symbols are characters, bytes, pixels, or audio samples—and then look at that symbol’s neighbors for some pattern or redundancy. This paper proposes an approach that examines the data stream in larger pieces, treats those pieces as arrays of bits where each row represents one symbol, and examines the arrays column by column rather than one row at a time.